Factor-Graph Aeromagnetic Compensation for Airborne Magnetic-Anomaly Navigation

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***Abstract*— Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft's own field must be compensated online when no calibration flight is available (the cold-start regime). A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We show that a factor-graph formulation reaches the same accuracy without a neural network by carrying the TL coefficients as graph variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight. Across four long lines from two flights and two maps, spanning free-flight navigation and survey missions, the smoother is the best of four cold-start estimators on every full-length line (8 of 8 cases): it stays bounded where a plain online-TL EKF and a marginalized particle filter both diverge, and it beats a reimplemented EKF+TL+NN, which the neural network keeps bounded, without using a neural network. A simulation Monte-Carlo confirms that the reported covariance is consistent and never over-confident. We also give an exact linear condition for the joint observability of compensation and navigation, from which an exogeneity design rule for the compensation features follows.**

***Index Terms*— Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.**

Nomenclature

$\mathbf{x}_t$	INS error state at epoch t (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$.
$\boldsymbol{\beta}_t$	Online Tolles–Lawson compensation coefficients.
$\boldsymbol{\chi}_t$	Joint state $(\mathbf{x}_t, \boldsymbol{\beta}_t)$.
$\boldsymbol{\Phi}_t, \mathbf{Q}_t$	Pinson state-transition matrix and process covariance.
z_t, η_t	Scalar magnetometer measurement and its noise, variance R.
$h(\cdot)$	Anomaly-map interpolation (with IGRF core) as a function of position.
$\mathbf{g}_t$	Map gradient $\partial h/\partial \mathbf{p}$ at epoch t.
$\mathbf{A}_t$	Tolles–Lawson basis row from the three-axis fluxgate.
c_t	Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$.
L_w, L_o	Sliding-window length and overlap (look-ahead).
ρ, δ	Robust kernel and its Huber threshold.
$\mathbf{G}, \boldsymbol{\Psi}$	Stacked map-gradient and TL-regressor matrices over a window.

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I. Introduction

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, and observable with a lightweight scalar magnetometer [3], [4]. It is also inherently hard to deny. Denial of a satellite signal exploits the far-field radio link, whose power falls off only as the inverse square of range, so a modest transmitter jams or spoofs a receiver from far away; a magnetic anomaly, by contrast, is a near-field source whose dipole field decays as the inverse cube of distance, so counterfeiting a credible anomaly at a moving aircraft would demand an impractically large or close field source [5]. The same map-matching idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [6], [7], [8]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft's own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The idea has matured

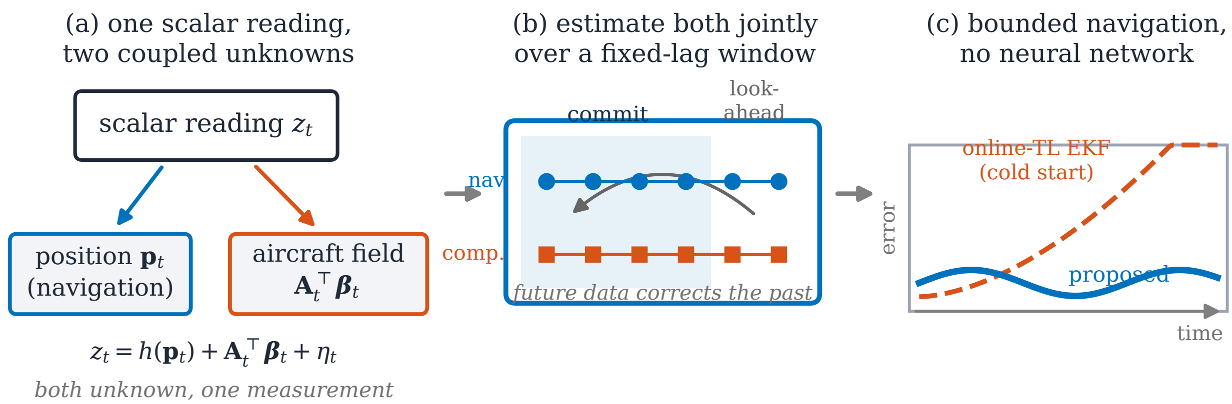

Fig. 1. Concept of the study: (a) one scalar reading couples aircraft position and aircraft-field compensation; (b) a fixed-lag factor graph estimates both jointly, letting later measurements correct earlier states; (c) bounded cold-start navigation, with no neural network.

from feasibility study to flight demonstration: recursive Bayesian map matching, from the extended Kalman filter to grid and particle estimators, was characterized on flight data [3], [4] and flown on an F-16 [9], and recent multi-hour fixed-wing trials over thousands of kilometres report metre-to-decametre accuracy with quantum magnetometers [5]. As the estimators have matured, attention has moved to the inputs they rely on, and a consensus is forming that the pacing requirement for operational deployment is a sufficiently accurate, uncertainty-aware anomaly map [10]. Two obstacles, in fact, separate these demonstrations from routine use, and they are distinct: the reference map on the ground and the aircraft's own field in the air. The map is a survey-and-data problem, pursued elsewhere [10]; this paper addresses the second.

The platform's own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position. This aircraft-field problem is intrinsic to MagNav and independent of any particular airframe; it is most acute for magnetometers mounted inside the cabin, close to avionics and current-carrying structure, rather than on a wingtip stinger boom that holds the sensor away from the platform field.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [11], [12], [13], a regression of the interference onto sixteen attitude-dependent terms (permanent, induced, and eddy-current) formed from a three-axis fluxgate magnetometer. It faces two difficulties that are easy to conflate but are in fact orthogonal. The first is *temporal*: the coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, yet many operational scenarios admit none, and the coefficients drift with temperature and configuration. In this cold-start regime they must be estimated online and jointly with navigation, from the same measurements that fix position, a coupling that makes both estimation and its analysis substantially harder than the calibrated case. The second is *representational*: however well its coefficients are fit, the linear TL basis cannot span interference from platform currents and other non-rigid, non-attitude sources, which contribute a large fraction of the field on cabin-mounted sensors [14], [15]. One difficulty is about identifying a known model class in time; the other is about that class being too small.

A public benchmark has made both concrete: the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset [14], with flight-grade INS, wingtip-stinger and cabin-mounted magnetometers, and high-resolution anomaly maps. Most work on it attacks the two difficulties *offline*: the TL coefficients by a batch calibration flight, and the representational gap by learned models fit offline on logged data—neural-network heading-error and denoising compensators [16], [15], physics-informed liquid-time-constant networks [17], and model-stability studies of the regression in gradiometry [18]. The exception most relevant here is the online EKF+TL+NN of [19], which carries an online-trained residual network inside the filter to mop up the un-modeled remainder, and is, to our knowledge, the strongest reported cold-start result on the benchmark. On top of the platform field, the scalar optically-pumped magnetometers used for the cabin channels add their own orientation-dependent errors—heading error and equatorial dead zones [20], [21]. This paper takes on the *temporal* difficulty fully—online, joint estimation of a time-varying compensation—and the physically structured part of the representational one, namely the scalar-sensor errors; the unstructured, current-driven remainder that a network is designed to absorb is left outside its scope.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [22] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [6] and nonlinear Kalman filtering for terrain-aided navigation [7] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [23], [24], [25], [8]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [9]. Beyond the EKF, unscented filtering [26], invariant formulations that respect the state manifold [27], and mixture/particle representations [28] improve consistency under nonlinearity. However, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [11], [12], [13]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [16], model-stability and robustness of the regression have been studied in gradiometry [18], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [15]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses [19]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and is, to our knowledge, the strongest reported cold-start result on that benchmark. Nevertheless, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and, being embedded in an EKF, inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [29] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [30], [31], the same structure underlying bundle adjustment and graph SLAM [32], [33], [34]. Incremental and square-root smoothers (square-root SAM [35], iSAM [36], iSAM2 [37], and exactly-sparse delayed-state filters [38]) make the solution efficient enough for online use, and fixed-lag sliding-window optimization [39], [40] bounds its cost. On-manifold inertial preintegration [41], factor-graph inertial fusion [42], and modern visual–inertial estimators [43] have brought the approach firmly into navigation. Despite these advances, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks. The factor graph has very recently reached aeromagnetic compensation itself: Lathrop et al. [44] calibrate a magnetometer by fusing vector and scalar magnetometers with inertial measurements in a factor graph, absorbing large permanent moments and a time-varying external field. That work compensates the sensor but does not close the loop to navigation. The joint problem, estimating a time-varying compensation together with aircraft position from map matching, has not been posed in a factor graph, and the observability of doing so has not been characterized.

Taken together, the three lines leave a specific gap. The cold-start MagNav problem is fundamentally a joint estimation of position and a time-varying compensation model from a single scalar stream, in which the two unknowns are confounded through the very measurement that must resolve them. Causal filters (cluster A) must commit each state before the maneuvers that would disambiguate the compensation have occurred, and so diverge when the initial residual is large; learned online compensation (cluster B) repairs that residual but keeps the causal, commit-online architecture and adds parameters that are difficult to certify; and factor-graph smoothing (cluster C), which can defer commitment and estimate structured nuisance variables jointly, has not been applied to this problem, nor has anyone characterized when the joint problem is even observable. What is missing is an estimator that (i) defers commitment so that late-arriving calibration information corrects early navigation states, (ii) represents the compensation (and the physical sensor errors of the scalar magnetometer) as first-class, time-varying variables rather than a black box, and (iii) comes with an explicit condition for when position and compensation can be separated at all. This paper supplies all three.

In this work we cast magnetic-anomaly navigation as a factor graph and, within it, carry the online Tolles–Lawson coefficients as time-varying variables, solving the joint compensation-and-navigation problem as a fixed-lag sliding-window smoother (Fig. 1). Making the compensation a set of graph variables that the window relinearizes is what lets it adapt along the flight, and deferring commitment over the window is what lets late-arriving calibration correct early navigation. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [45], so smoothing per se is not the contribution—the joint, windowed estimation of a time-varying compensation is. The paper makes the following contributions:

1) **Factor-graph MagNav with joint online compensation.** We give, to our knowledge, the first formulation of magnetic-anomaly navigation as factor-graph (batch MAP) estimation, and carry the online Tolles–Lawson coefficients as time-varying variables of a fixed-lag sliding-window graph so that navigation and compensation are estimated jointly. The per-window relinearization makes the compensation adapt along the flight—the role a learned online model plays in the competing

filter—while the graph keeps every quantity an explicit, physically interpretable state (Section III). We are deliberate about what is claimed: for the linear–Gaussian error model the batch estimate equals the RTS smoother, so smoothing per se is not the contribution; the joint, windowed estimation of a time-varying compensation is.

2) **Exact joint-observability condition.** We prove a necessary and sufficient linear-algebraic condition under which the aircraft position and the compensation coefficients are jointly observable from the scalar magnetometer, separating three distinct failure modes (an unexcited trajectory, a rank-deficient compensation basis, and the subtle one, a compensation subspace that is confounded with the anomaly gradient). From it we derive a concrete exogeneity design rule for choosing compensation features that a filter tuned by trial and error cannot see (Section IV).
3) **Physical sensor-error factors.** We add physically motivated optically-pumped-magnetometer error terms: a harmonic heading error carried as extra graph variables, and a dead-zone attenuation handled as a heteroscedastic reweighting of the scalar measurement. A controlled cumulative ablation then shows which of them are load-bearing. These are model terms that a position-only estimator cannot represent, and they let the graph absorb structured sensor error that would otherwise leak into the position estimate (Section III, Table V).
4) **Cold-start capability, validated against a reproduced baseline.** Deferring commitment over the window lets the estimator stay bounded from a cold start with no calibration flight and no learned component. To measure this fairly we reimplement, rather than merely cite, the online EKF+TL+NN of [19] and compare it, a weak online-TL EKF, and a marginalized particle filter carrying the TL coefficients against the proposed smoother on the uncompensated cabin magnetometers, across eight cold-start cases spanning four lines, two flights, and two maps. The comparison isolates what the network buys (it keeps the causal filter bounded where online TL alone and the particle filter diverge) from what it does not (on every full-length line the smoother is the most accurate of the four), a like-for-like evaluation the literature has lacked (Section VI).

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the sliding-window smoother and the compensation and sensor-error factors. Section IV establishes the observability condition. Section V describes the reimplemented EKF+TL+NN baseline, and Section VI reports and interprets the experiments. Section VII concludes and states the limitations.

II. Problem Formulation

We first fix notation and state the estimation problem. Throughout, boldface lowercase denotes vectors and boldface uppercase matrices; $\mathcal{N}(\cdot,\cdot)$ is the Gaussian density; and $\|\mathbf{v}\|^2_{\mathbf{M}} = \mathbf{v}^\top \mathbf{M} \mathbf{v}$ is the squared Mahalanobis norm.

A. INS error state and dynamics

We adopt the standard Pinson error model of a strapdown INS [2], [1], [46]. The error state

$$\mathbf{x} = \left[\, \delta\mathbf{p}^\top \; \delta\mathbf{v}^\top \; \boldsymbol{\psi}^\top \; h_a \; \hat{a} \; \mathbf{b}_a^\top \; \mathbf{b}_g^\top \; S \,\right]^\top \in \mathbb{R}^{18} \tag{1}$$

stacks position error $\delta\mathbf{p}$, velocity error $\delta\mathbf{v}$, attitude error $\boldsymbol{\psi}$ (three components each), the baro-inertial altitude-aiding pair $(h_a, \hat{a})$ that damps the vertically unstable channel, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. The discretized, linearized dynamics are

$$\mathbf{x}_{t+1} = \boldsymbol{\Phi}_t \mathbf{x}_t + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \tag{2}$$

where $\boldsymbol{\Phi}_t$ is the Pinson state-transition matrix (Appendix A) and $\mathbf{Q}_t$ the process covariance; the FOGM block of $\boldsymbol{\Phi}_t$ is $e^{-\Delta t/\tau}$ with correlation time τ.

B. Scalar magnetometer measurement

The scalar magnetometer reads

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R), \tag{3}$$

where $h(\cdot)$ interpolates the crustal anomaly map (with the IGRF core field restored) at the aircraft position $\mathbf{p}_t = \bar{\mathbf{p}}_t + \delta\mathbf{p}_t$ and c_t is the aircraft interference. Linearizing about the nominal position,

$$z_t \approx h(\bar{\mathbf{p}}_t) + \mathbf{g}_t^\top \delta\mathbf{p}_t + c_t + \eta_t, \qquad \mathbf{g}_t = \left.\frac{\partial h}{\partial \mathbf{p}}\right|_{\bar{\mathbf{p}}_t}, \tag{4}$$

so the map gradient $\mathbf{g}_t$ is the sole source of horizontal position information: where $\mathbf{g}_t \approx \mathbf{0}$ (a locally flat map) position is momentarily unobservable, and everywhere the interference c_t competes with the $\mathbf{g}_t^\top \delta\mathbf{p}_t$ term for the same measurement. The map h is stored as a gridded anomaly product upward-continued to the survey altitude; we evaluate it and its gradient by bicubic interpolation, which yields the continuous $\mathbf{g}_t$ required to linearize (4), and we restore the IGRF core field so that z_t and h are on a common absolute scale. The gradient magnitude $\|\mathbf{g}_t\|$ varies by more than an order of magnitude along a line, which is why position accuracy is not uniform in time even for a perfect compensator, a point the error traces of Section VI make visible.

C. Tolles–Lawson interference model

Classical aeromagnetic compensation writes the interference as a linear combination of attitude-dependent basis terms [11], [12], [13],

$$c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t, \tag{5}$$

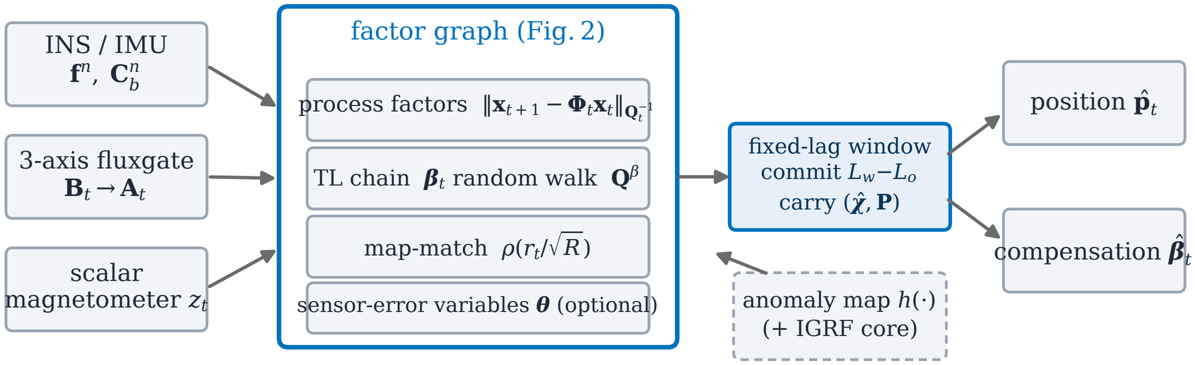

Fig. 2. Proposed estimator: the sensors feed a factor graph of navigation, online TL compensation, and optional sensor-error variables, solved over a fixed-lag window.

where $\mathbf{A}_t \in \mathbb{R}^m$ is the TL basis row of (5) built from the three-axis fluxgate; its classical rank deficiency is a gauge freedom (Remark 2). With direction cosines $\hat{\mathbf{b}}_t = [\,u\ v\ w\,]^\top$ of the Earth field in body axes and total field magnitude B_t, the permanent (3), induced (6), and eddy-current (9) groups are

$$\mathbf{A}_t = \big[\underbrace{u, v, w}_{\text{perm.}},\ \underbrace{B_t u^2, \dots, B_t w^2}_{\text{induced}},\ \underbrace{B_t u\dot{u}, \dots, B_t w\dot{w}}_{\text{eddy}}\big]^\top, \tag{6}$$

optionally augmented by a bias term, giving $m \le 19$ coefficients $\boldsymbol{\beta}_t$. In the calibrated setting $\boldsymbol{\beta}$ is fit once and frozen; in the cold-start setting it is unknown at takeoff and drifts slowly, so we treat it as a time-varying state with a random-walk prior $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{w}_t^\beta$, $\mathbf{w}_t^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta)$.

D. Joint estimation problem

The cold-start setting couples the two preceding models through a single scalar stream: the same measurement z_t must simultaneously fix position through the map term $h(\mathbf{p}_t)$ and identify the compensation $\mathbf{A}_t^\top \boldsymbol{\beta}_t$. We therefore state the estimation task as one joint problem rather than a navigation problem preceded by a calibration step.

Problem 1 (Joint compensation and navigation): Given the scalar measurements $z_{1:N}$ of (3), the fluxgate regressor rows $\mathbf{A}_{1:N}$ of (6), the anomaly map h, the INS mechanization entering $\mathbf{\Phi}_t$, and the prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$, estimate the joint trajectory $\boldsymbol{\chi}_{1:N}$ of navigation errors and compensation coefficients, $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$.

We take the maximum-a-posteriori (MAP) estimate. By the Markov structure of the dynamics (2), the coefficient random walk, and the memoryless measurement (3), the joint posterior factorizes as

$$p(\boldsymbol{\chi}_{1:N}\,|\,z_{1:N}) \propto p(\boldsymbol{\chi}_1)\prod_{t=1}^{N-1} p(\mathbf{x}_{t+1}\,|\,\mathbf{x}_t)\, p(\boldsymbol{\beta}_{t+1}\,|\,\boldsymbol{\beta}_t)\prod_{t=1}^{N} p(z_t\,|\,\boldsymbol{\chi}_t). \tag{7}$$

This factorization *is* a factor graph [29], [30]: a bipartite graph whose variable nodes are the $\boldsymbol{\chi}_t$ and whose factor nodes are the individual densities in (7), each connected only to the variables it involves (here, two parallel chains coupled by the measurement factors; Fig. 3). MAP estimation on such a graph is sparse nonlinear least squares: the negative logarithm of (7) turns each factor into a squared—or robust—residual, and the Gauss–Newton normal equations inherit the graph's chain sparsity, which is what makes batch solution tractable at flight scale (Section III-E). Concretely, the MAP estimate minimizes

$$J(\boldsymbol{\chi}) = \|\boldsymbol{\chi}_1 - \boldsymbol{\chi}_0\|^2_{\mathbf{P}_0^{-1}} + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \mathbf{\Phi}_t \mathbf{x}_t\|^2_{\mathbf{Q}_t^{-1}} + \sum_{t=1}^{N-1} \|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}} + \sum_{t=1}^{N} \rho\left(\frac{r_t(\boldsymbol{\chi}_t)}{\sqrt{R}}\right), \tag{8}$$

with measurement residual $r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t$ and ρ a possibly robust kernel; the prior term anchors both blocks of $\boldsymbol{\chi}_1 = (\mathbf{x}_1, \boldsymbol{\beta}_1)$, at cold start with $\boldsymbol{\beta}$-block $\mathbf{P}_0^\beta$ (Table I). We make the standard modeling assumptions of aided inertial navigation [1], [28].

Assumption 1: The process and measurement noises $\mathbf{w}_t, \mathbf{w}_t^\beta, \eta_t$ are zero-mean, white, mutually independent, and Gaussian; the linearizations (2) and (4) are valid over a window; and the prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$ is Gaussian.

Assumption 1 is the same one under which the EKF and RTS smoother are derived, so any performance difference between the estimators reflects their structure, not their noise models. What Problem 1 adds over classical map-matching is exactly the second chain: navigation and compensation are estimated from the same evidence, and Section IV characterizes when that joint problem is well-posed.

III. Method

The proposed estimator (Fig. 2) represents navigation, online compensation, and optional sensor-error states as variables of a single factor graph, and solves it over a fixed-lag sliding window by robust Gauss–Newton. This section develops the factors, the windowing, the robust solver, and the sensor-error terms.

A. Online TL coefficients as graph variables

We append the compensation coefficients $\boldsymbol{\beta}$ to the state, so the measurement factor of (3) becomes, in residual form,

$$r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t, \tag{9}$$

with Gaussian information R^{-1}, and its Jacobian row is

$$\mathbf{H}_t = \Big[\underbrace{\mathbf{g}_t^\top}_{\delta\mathbf{p}} \quad \underbrace{\mathbf{0}}_{\delta\mathbf{v},\boldsymbol{\psi},h_a,\hat{a},\mathbf{b}} \quad \underbrace{\mathbf{A}_t^\top}_{\boldsymbol{\beta}} \quad \underbrace{1}_{S}\Big]. \tag{10}$$

The window posterior over the joint states $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$ is the product of four factor families, drawn explicitly in Fig. 3:

1) a joint prior on $\boldsymbol{\chi}_1$: $\mathbf{P}_0$ at cold start, or the smoothed mean and covariance carried from the previous window (Section III-B);
2) Pinson–FOGM process factors $\|\mathbf{x}_{t+1} - \boldsymbol{\Phi}_t\mathbf{x}_t\|^2_{\mathbf{Q}_t^{-1}}$, parameterized by the INS mechanization (specific force $\mathbf{f}^n$, attitude $\mathbf{C}_b^n$) through (13);
3) compensation random-walk factors $\|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}}$, which set how quickly the compensation may adapt:$\mathbf{Q}^\beta \to \mathbf{0}$ recovers a single static $\boldsymbol{\beta}$;
4) robust scalar-magnetometer factors $\rho(r_t/\sqrt{R})$ with the residual (9), each coupling $\delta\mathbf{p}_t$ (through the map gradient $\mathbf{g}_t$), the disturbance state S_t, and $\boldsymbol{\beta}_t$ (through the fluxgate regressor $\mathbf{A}_t$).

The INS mechanization, the fluxgate (through $\mathbf{A}_t$), and the anomaly map (through h and $\mathbf{g}_t$) enter as known factor parameters, not as estimated variables; the optional sensor-error variables $\boldsymbol{\theta}$ of Section III-F attach to every measurement factor when enabled. Two implementation details: in the online estimator only the horizontal components of $\mathbf{g}_t$ are used (the vertical channel is baro-damped and the map is evaluated at survey altitude), whereas the batch validation of Section VI-A uses the full three-component gradient; and the implementation orders the state vector as [Pinson-17; $\boldsymbol{\beta}$; S], a permutation of (1) that does not affect the estimate. Because consecutive coefficient blocks are chained rather than instantaneous, the compensation is identified from the whole trajectory, and because the batch solution uses future as well as past measurements, early navigation states benefit from calibration information that only becomes observable after later maneuvers, the property a causal filter lacks.

It is worth being precise about the causal failure this repairs, because the same time-varying model (5) can be, and is, carried inside an EKF. At cold start the coefficient error makes the residual hundreds of nanotesla, and a one-pass filter must apportion that residual between position and $\boldsymbol{\beta}$ immediately, using only its current covariance, before the maneuvers that would separate the two have occurred. A wrong apportionment moves the position estimate, and the next update then evaluates the map gradient $\mathbf{g}_t$ at the wrong place, so even a correct residual is charged to the wrong direction: the linearization point and the estimate corrupt each other, and because the filter cannot revise committed states, the loop runs open until divergence. Innovation gating does not repair this, because the cold-start residual is not an outlier but a systematic bias present in every early measurement: a chi-square gate rejects essentially all of them, which removes the only evidence from which $\boldsymbol{\beta}$ could be identified and leaves free-inertial drift. The window changes both halves of the failure. Each Gauss–Newton iteration relinearizes every epoch in the window, so once the coefficients begin to converge mid-window, the residual attribution and the linearization points of the *early* epochs are revised retroactively; and the robust kernel, unlike a gate, keeps every measurement, merely down-weighting what the current model cannot yet explain. The breadth study isolates the two effects: the window alone, with no robust kernel, already stays bounded on every counted case (Section VI-D), so the load-bearing mechanism is the windowed relinearization, with the kernel as an accuracy refinement.

Absorbing the non-Tolles–Lawson remainder. On this dataset the cabin interference is dominated by platform currents and other sources the attitude-only TL basis $\mathbf{A}_t$ cannot span, and the online EKF+TL+NN of [19] devotes a neural network precisely to this remainder. The proposed graph carries no network; instead three of the factors above combine to absorb the remainder as a structured, robust regression problem. The random-walk factor (iii) makes $\boldsymbol{\beta}_t$ time-varying: writing the true interference as $\mathbf{A}_t^\top\boldsymbol{\beta}^\star + e_t$, a static fit ($\mathbf{Q}^\beta \to \mathbf{0}$) forces the entire current signature into the residual e_t, whereas a nonzero $\mathbf{Q}^\beta$ lets the estimated coefficients track the slowly drifting, maneuver-correlated projection of the currents onto $\mathbf{A}_t$, the low-frequency part a single calibration fit cannot recover. The disturbance state S_t in the residual (9), a first-order Gauss–Markov process folded into the Pinson–FOGM factor (ii), then carries the temporally correlated component of e_t that is orthogonal to $\mathbf{A}_t$; it is the linear–Gaussian, single-scalar analogue of the learned residual, cheaper and far less expressive but requiring no training data. What neither term explains, the heavy-tailed switching transients of onboard electronics, is handled not by fitting but by the robust kernel in factor (iv), which the reweighting of Section III-D down-weights so that such samples cannot pull the position estimate. The observability condition of Section IV is what keeps this well-posed: S_t and $\boldsymbol{\beta}_t$ may only absorb signal that is not collinear with the map gradient $\mathbf{g}_t$, so genuine map information is not silently explained away as interference. The trade it makes, a higher compensation residual than the network in exchange for navigation robustness, is quantified with the cold-start results in Section VI-C, and the account itself is checked against the recorded platform-current channels in Section VI-D.

B. Fixed-lag sliding window

A single batch over a long line forces one static $\boldsymbol{\beta}$, which underfits time-varying interference; a per-epoch filter, conversely, cannot use future data. We take the middle path and process the flight in overlapping windows of length L_w with overlap L_o, each a batch factor graph, in the manner of fixed-lag and incremental smoothing [37], [39], [40]. Window k spans epochs $[s_k, s_k + L_w)$; it is solved as a batch, commits only its leading stride of $L_w - L_o$ epochs, and carries the full smoothed state and covariance of the last committed epoch forward as a Gaussian prior for window $k{+}1$ (Fig. 4). The trailing overlap L_o acts as bounded smoother look-ahead, so each committed estimate has seen L_o epochs of future data. Carrying the full covariance, not just the mean, keeps navigation continuous across window boundaries. One approximation deserves note: the carried prior is the smoothed state at the commit boundary, which has already seen the overlap measurements, and window $k{+}1$ processes those L_o epochs again, so the overlap information is counted twice. An exactly consistent scheme would carry the forward-filtered state or marginalize the overlap; we accept the bounded optimism (the overlap is a fixed fraction, $L_o/L_w = 0.3$, of each window) in exchange for the simpler implementation, and note that the reported DRMS is a comparison of means, unaffected by the covariance bookkeeping. Algorithm 1 summarizes the procedure; its per-window cost is constant in the line length, giving overall linear complexity.

Algorithm 1 Fixed-lag sliding-window smoother

Input: measurements $\{z_t\}$, TL rows $\{\mathbf{A}_t\}$, map h, window L_w, overlap L_o, prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$
$s \leftarrow 1$
while $s \le T$ **do**
  build factor graph on epochs $[s, \min(s{+}L_w, T))$ with prior $(\hat{\boldsymbol{\chi}}_{s-1}, \mathbf{P}_{s-1})$
  repeat ▷ robust Gauss–Newton
    linearize residuals (9); form weights w_t from ρ (12)
    solve the linearized chain by an iterated RTS pass [45] (equivalently, sparse QR / square-root SAM [35]; Sec. III-E)
    update $\boldsymbol{\chi} \leftarrow \boldsymbol{\chi} + \Delta\boldsymbol{\chi}$
  until converged
  commit epochs $[s, s{+}L_w{-}L_o)$; recover $(\hat{\boldsymbol{\chi}}, \mathbf{P})$ at epoch $s{+}L_w{-}L_o{-}1$
  $s \leftarrow s + (L_w - L_o)$
end while
Output: committed trajectory $\{\hat{\mathbf{p}}_t, \hat{\boldsymbol{\beta}}_t\}$

C. Window design and initialization

Two parameters govern the window: the length L_w and the overlap L_o. L_w sets how much data each compensation estimate sees and thus the effective adaptation time; too short and $\boldsymbol{\beta}$ is under-determined, too long and it cannot track drift. L_o sets the look-ahead and hence the commit latency: a committed state has seen L_o epochs of future data, and the estimator lags real time by L_o. In the experiments we use $L_w = 5\,\mathrm{min}$ and $L_o = 90\,\mathrm{s}$; the $2\,\mathrm{min}$ window in Table II shows that shortening L_w degrades the long-line result, consistent with under-determination. The first window is seeded with a diffuse prior on $\boldsymbol{\beta}$ (large $\mathbf{P}_0$ block) so the cold start is not biased toward zero interference, and the FOGM disturbance state absorbs the initial mismatch while the coefficients converge; subsequent windows inherit the tight carried covariance and converge in a few Gauss–Newton iterations (typically 3–5; Sec. VI-F). This warm-start is why the per-window iteration count stays low despite the nonlinearity of the map term.

D. Robust kernel and reweighting

Map error, transient interference, and dead-zone measurements produce heavy-tailed residuals that a quadratic cost over-weights. We use a Huber kernel [47]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \delta, \\ \delta\big(|u| - \frac{1}{2}\delta\big), & |u| > \delta, \end{cases} \tag{11}$$

minimized by iteratively reweighted least squares (IRLS): each Gauss–Newton iteration reweights measurement factor t by

$$w_t = \min\big(1,\ \delta/|u_t|\big), \qquad u_t = r_t/\sqrt{R}, \tag{12}$$

so gross outliers are down-weighted toward an absolute-value penalty while inliers keep the efficient quadratic cost. In the implementation the reweighting is applied from the second Gauss–Newton iteration onward (the first iteration is plain least squares) with w_t clamped to $[10^{-6}, 1]$, and $\delta = 1.345$ gives the standard 95% Gaussian efficiency. Alternatives such as switchable constraints [48] and dynamic covariance scaling [49] fit the same graph interface and could replace (11) unchanged.

E. Sparse solver

Each Gauss–Newton iteration solves a sparse linear system whose Jacobian has the block-bidiagonal structure of the two chains in Fig. 3 (in the static-$\boldsymbol{\beta}$ limit the coefficient blocks collapse to one shared column). We factor it by column-pivoted QR on the square-root information matrix (square-root SAM [35], [30]), which is numerically stable and exploits the chain sparsity. An iterated RTS pass [45] gives the identical estimate

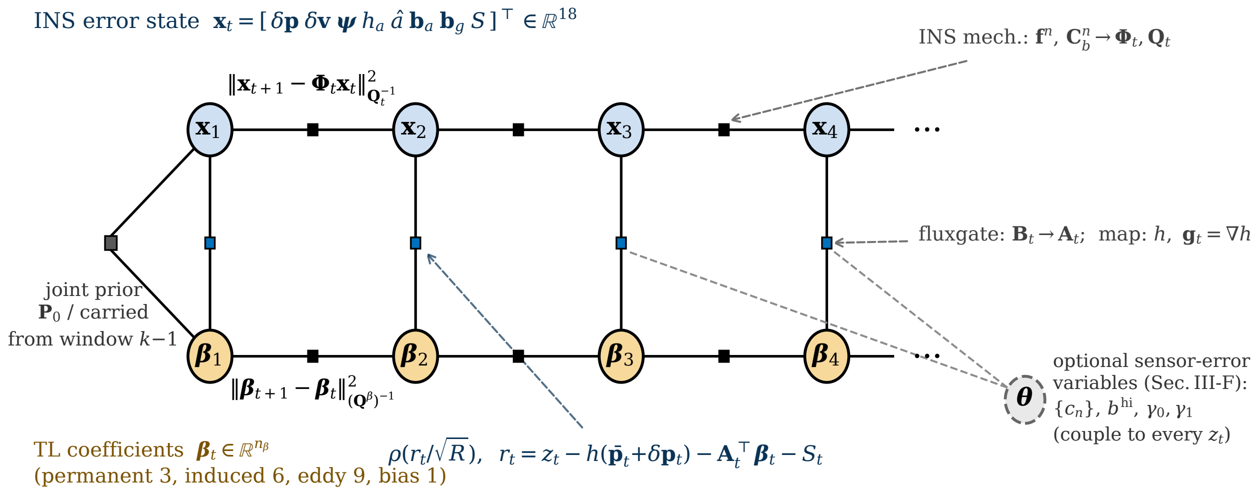

Fig. 3. Factor graph of one window: the 18-state INS error chain $\mathbf{x}_t$ and the time-varying TL chain $\boldsymbol{\beta}_t$, coupled by robust scalar-magnetometer factors. Dashed arrows are known factor parameters; the dashed node $\boldsymbol{\theta}$ holds the optional sensor-error variables.

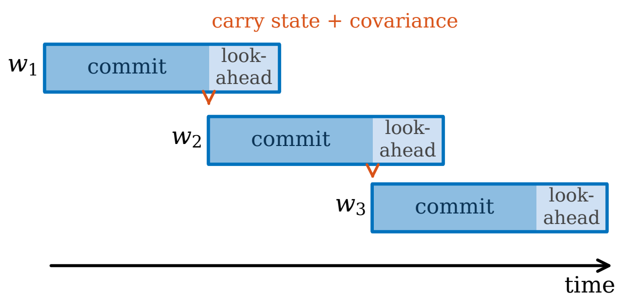

Fig. 4. Fixed-lag sliding window: each window commits its leading stride and carries the smoothed state and covariance to the next; the overlap is the look-ahead.

in the Gaussian case; the equivalence is exact, not an approximation:

Lemma 1:
Under Assumption 1, with $\rho(u) = \frac{1}{2}u^2$ and $\boldsymbol{\beta}$ held fixed, the minimizer of (8) equals the Rauch–Tung–Striebel (RTS) smoother estimate over the window.

Proof:
With ρ quadratic and $\boldsymbol{\beta}$ fixed, (8) is a positive-definite quadratic in the Gaussian chain $\{\mathbf{x}_t\}$, so its minimizer is the posterior mean $\mathbb{E}[\mathbf{x}_{1:N} \mid z_{1:N}]$. For a linear–Gaussian state-space model that conditional mean is exactly what the forward Kalman recursion followed by the backward RTS recursion computes [45]; the two therefore coincide. ■

Lemma 1 does double duty. Practically, it licenses the fast iterated-RTS implementation used for the joint TL problem, and we verify the QR and RTS paths agree to within a metre on a real-data segment in Section VI-A (that check covers the navigation-only chain). Conceptually, it fixes the scope of the contribution: for a static, known compensation the batch graph and a filter–smoother pass are equivalent, so nothing here rests on "smoothing beats filtering." What the graph adds beyond Lemma 1 is exactly what the lemma's hypotheses exclude: the jointly estimated, time-varying $\boldsymbol{\beta}$, the non-Gaussian robust kernel of (11), the sensor-error variables of Section III-F, and the fixed-lag windowing that keeps latency bounded.

F. Physical sensor-error factors

The graph representation admits error states that a position-only grid or point-mass estimator cannot carry. Beyond TL we optionally augment χ with three physically motivated scalar-magnetometer error variables, each appearing additively in the residual (9), plus one measurement re-weighting:

1) OPM heading error, modeled as the cosine-harmonic expansion of the optically-pumped (scalar-atomic) magnetometer in the sensor–field angle $\psi_t = \angle(\hat{\mathbf{e}}, \mathbf{B}_t)$, $b_t^{\mathrm{hdg}} = \sum_{n\in\{1,2,4\}} c_n \cos(n\psi_t)$ [20], [21], with coefficients $\{c_n\}$ as graph variables; the $n{=}1$ term is the vector light shift and $n{=}2$ the nonlinear Zeeman effect. Because ψ_t is known from attitude and IGRF, the term is linear in $\{c_n\}$ and, as in the Tolles–Lawson case [21], enters the graph as an ordinary linear regressor;
2) Hard-iron bias b^{hi}, a constant offset variable;
3) Linear drift $b_t^{\mathrm{dr}} = \gamma_0 + \gamma_1 t$, capturing slow electronics drift;
4) Dead-zone weight: not a graph variable but a heteroscedastic measurement information $R_t^{-1} = R^{-1}\max(|\cos\psi_t|, \epsilon)^2$ ($\epsilon{=}0.05$) that follows the OPM signal-amplitude collapse near the equatorial dead zone (field $\perp$ optical axis) [20] and de-weights those samples (a de-weighting any esti-

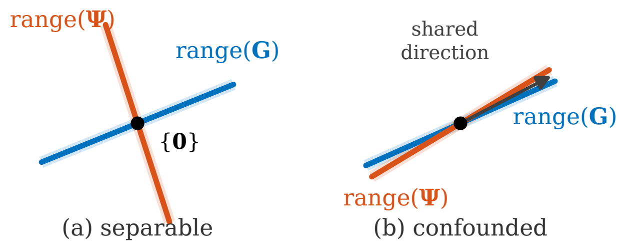

Fig. 5. Observability geometry (Proposition 1): position and compensation are separable when $\mathrm{range}(\mathbf{G})$ and $\mathrm{range}(\boldsymbol{\Psi})$ meet only at the origin (left) and confounded when they share a direction (right).

mator could apply; it is included for completeness of the sensor model).

The augmented residual is $r_t = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t - b_t^{\mathrm{hdg}} - b^{\mathrm{hi}} - b_t^{\mathrm{dr}}$. Each term is exogenous to position (Section IV), so it can be identified without stealing map information; their observability is limited only by how much the flight excites the corresponding regressor (e.g. the heading harmonics need attitude variation).

IV. Observability of Joint Compensation and Navigation

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let $\mathbf{G}$ have rows $\mathbf{g}_t^\top$ (the map-gradient/position sensitivity) and let $\boldsymbol{\Psi}$ have rows $\mathbf{A}_t^\top$ (the TL regressor). To isolate the position–compensation coupling, restrict (3) to the $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ subspace, giving the window model $\mathbf{z} = \mathbf{G}\,\delta\mathbf{p} + \boldsymbol{\Psi}\,\delta\boldsymbol{\beta} + \boldsymbol{\eta}$ (Fig. 5). This model freezes $\delta\mathbf{p}$ and $\delta\boldsymbol{\beta}$ over the window: it is a per-window surrogate for the time-varying system (2), (9) that isolates the position–compensation coupling which a full observability Gramian would mix with the INS dynamics.

Proposition 1:
A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment, hence unobservable from map matching over the window, if and only if $\mathbf{G}\,\delta\mathbf{p} \in \mathrm{range}(\boldsymbol{\Psi})$. Consequently the joint state $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ is observable if and only if the stacked matrix $[\,\mathbf{G} \quad \boldsymbol{\Psi}\,]$ has full column rank; equivalently, if and only if all three of the following hold: (i) $\ker \mathbf{G} = \{\mathbf{0}\}$ (the map gradient excites every position direction over the window); (ii) $\ker \boldsymbol{\Psi} = \{\mathbf{0}\}$ (the compensation basis has full column rank); and (iii) $\mathrm{range}(\mathbf{G}) \cap \mathrm{range}(\boldsymbol{\Psi}) = \{\mathbf{0}\}$ (no shared direction). Condition (iii) alone governs the position–compensation confounding: when it fails, a position shift with $\mathbf{G}\,\delta\mathbf{p} \neq \mathbf{0}$ is reproduced exactly by a compensation change.

Proof:
Two joint states $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ and $(\delta\mathbf{p}', \delta\boldsymbol{\beta}')$ produce identical noise-free predictions iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \boldsymbol{\Psi}(\delta\boldsymbol{\beta}' - \delta\boldsymbol{\beta})$, i.e. iff $[\,\mathbf{G} \quad \boldsymbol{\Psi}\,][(\delta\mathbf{p} - \delta\mathbf{p}')^\top \;\; (\delta\boldsymbol{\beta} - \delta\boldsymbol{\beta}')^\top]^\top = \mathbf{0}$. The unobservable subspace is therefore exactly $\ker[\,\mathbf{G} \quad \boldsymbol{\Psi}\,]$, and the joint state is observable iff $[\,\mathbf{G} \quad \boldsymbol{\Psi}\,]$ has full column rank; the first statement follows by taking $\delta\boldsymbol{\beta}'$ free and $\delta\mathbf{p}' = \mathbf{0}$. For the three-condition equivalence: if (i) fails, take $\delta\mathbf{p} \in \ker \mathbf{G} \setminus \{\mathbf{0}\}$, giving the kernel element $(\delta\mathbf{p}, \mathbf{0})$; if (ii) fails, take $\delta\boldsymbol{\beta} \in \ker \boldsymbol{\Psi} \setminus \{\mathbf{0}\}$, giving $(\mathbf{0}, \delta\boldsymbol{\beta})$; if (iii) fails, take $\mathbf{G}\,\delta\mathbf{p} = \boldsymbol{\Psi}\,\delta\boldsymbol{\beta} \neq \mathbf{0}$, giving $(\delta\mathbf{p}, -\delta\boldsymbol{\beta})$. Conversely, let $(\delta\mathbf{p}, \delta\boldsymbol{\beta}) \neq (\mathbf{0}, \mathbf{0})$ lie in the kernel, so $\mathbf{G}\,\delta\mathbf{p} = -\boldsymbol{\Psi}\,\delta\boldsymbol{\beta}$. If $\mathbf{G}\,\delta\mathbf{p} = \mathbf{0}$ then either $\delta\mathbf{p} \neq \mathbf{0}$ violates (i) or $\delta\boldsymbol{\beta} \neq \mathbf{0}$ violates (ii); if $\mathbf{G}\,\delta\mathbf{p} \neq \mathbf{0}$ it is a nonzero shared direction, violating (iii). ■

Corollary 1 (Exogeneity):
Suppose every compensation feature is exogenous: $\mathbf{A}_t$ is a function of aircraft attitude and the fluxgate alone, so $\boldsymbol{\Psi}$ does not depend on the estimated position. Then two conditions are necessary for (iii): the window must satisfy $N \geq \mathrm{rank}(\mathbf{G}) + \mathrm{rank}(\boldsymbol{\Psi})$, and no $\boldsymbol{\Psi}$ column may be a linear combination of the map-gradient time series. In particular, enlarging an exogenous basis can slow the coefficient estimate through (ii) but creates no structural coupling to position.

For an exogenous basis, condition (iii) can fail only through coincidence of the flown attitude history with the local map gradient—plausible, for instance, when turns always occur at the same map features—and one expects it to hold for trajectories and maps in general position once the dimension count above is met. We state this as an expectation, not a theorem: formalizing "general position" for real flight paths is beyond this paper, and Section VI shows that scalar proxies for the intersection are unreliable in practice. The dependable rule is the structural one: keep the basis exogenous.

Remark 1 (Endogenous features cancel the position channel):

An endogenous feature breaks the model's premise rather than condition (iii). If a compensation feature is itself a function of position—the measured scalar z_t is the canonical example—then $\boldsymbol{\Psi}$ is no longer a known parameter, and the linearization must charge the feature's own position sensitivity to the state. For the scalar reading that sensitivity is exactly the map gradient, $\partial z_t / \partial \mathbf{p} = \mathbf{g}_t$, so with fitted coefficient β_z on that feature the effective position row of the measurement becomes $(1 - \beta_z)\,\mathbf{g}_t$: as the regression drives $\beta_z \to 1$, the fit residual shrinks while the measurement loses its position information entirely. This is the collapse observed empirically in Section V, where the residual stays small as the position drifts to kilometres, and it is why the scalar magnetometer must be excluded from any compensation basis, learned or linear.

Remark 2 (TL rank deficiency as gauge freedom):
Condition (ii) fails, or very nearly fails, for the classical TL basis itself: with unit direction cosines the three induced diagonal columns sum to $B_t(u^2 + v^2 + w^2) = B_t$, which is collinear with the constant-bias column to the degree that the total field is constant over a window

(B_t varies by well under a percent of its $5 \times 10^4\,\mathrm{nT}$ magnitude), the long-known collinearity of TL regression [12], [18]. This is, to that approximation, a gauge freedom of the compensation rather than a navigation defect: a perturbation $\delta\boldsymbol{\beta}$ in the near-null subspace of $\boldsymbol{\Psi}$ moves the predicted interference $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$, and hence the position estimate, only negligibly. The prior $\mathbf{P}_0$ and the random-walk factors select one representative of each near-gauge orbit. Separability of position from compensation is governed by (iii) alone and is unaffected.

Proposition 1 is exact but non-constructive: it does not yield a reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection (in-sample basis R^2, out-of-sample prediction, and gradient-alignment indices) did not generalize across estimators or lines. The actionable consequence is Corollary 1: drive the compensation from exogenous features, which every estimator in Section VI does.

Remark 3:
The condition is geometric, not statistical: it depends on the ranges of $\mathbf{G}$ and $\boldsymbol{\Psi}$ over the window, hence on the flight path and the local map gradient, not on noise levels. It explains two failure modes seen in practice. First, over a locally flat map $\mathbf{G} \approx \mathbf{0}$ and condition (i) fails: position is unobservable regardless of the compensation, a map-information floor that bounds every estimator, and that we observe on the shortest line in Section VI. Second, an endogenous feature cancels the position channel by the mechanism of Remark 1, so the compensator silently absorbs the map signal; excluding such features (Section V) is necessary for the baseline to converge.

V. Reimplemented Online EKF+TL+NN Baseline

To compare against [19] on equal footing we reimplement, rather than merely cite, its online EKF+TL+NN cold start. The filter augments the Pinson error state with the online TL coefficients $\boldsymbol{\beta}$ and the weights of a small feed-forward network, all propagated as EKF states and updated online against (3). Since neither the network architecture nor the training design of [19] is released, ours reproduces the published design principles rather than the exact filter.

A naive port diverged, and three fixes, each addressing an independently diagnosed failure, were needed. (i) We remove the large constant offset from the regression target and initialize the network output bias to it, because the raw offset ($-1724\,\mathrm{nT}$ on Mag 4, $+310\,\mathrm{nT}$ on Mag 5) otherwise drives the network bias to $\sim 5 \times 10^4\,\mathrm{nT}$ and the filter to `NaN`. (ii) We restrict the network inputs to the attitude-derived TL basis, because feeding the scalar magnetometer creates the endogenous feature of the Remark above and collapses horizontal position (Corollary 1). (iii) We inflate the measurement variance and the coefficient process noise through the warm-up, when the compensation is still grossly wrong, then relax them.

The reimplemented filter carries the permanent TL group and a single-hidden-layer network of eight units on that group (41 network-weight states), with a FOGM $(\sigma, \tau) = (3\,\mathrm{nT}, 180\,\mathrm{s})$ and post-warm-up measurement variance $12^2\,\mathrm{nT}^2$. It reaches 40.0/$17.5\,\mathrm{m}$ on Mag 4/5 (Table II): inside the reported $37\,\mathrm{m}$ to $58\,\mathrm{m}$ band on Mag 4, and $2.5\,\mathrm{m}$ to $3.5\,\mathrm{m}$ above the reported $14\,\mathrm{m}$ to $15\,\mathrm{m}$ on Mag 5. This is a functioning but not exact reimplementation, which we use as the baseline.

VI. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [14]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically-pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines accordingly. Flt1007 is the free-flight mission, in which the aircraft flies freely within the mapped Renfrew ($\approx 400\,\mathrm{m}$) and Eastern/Perth ($\approx 800\,\mathrm{m}$) areas, so its lines are the closest to a navigation task and are what prior work [14], [4], [19] reports; 1007.06 is our primary line for that reason. Flt1003 is a crustal-field measurement flight (survey traverses at 400 and $800\,\mathrm{m}$); its lines 1003.02 and 1003.08 are straighter than the free-flight legs. Flt1006 is a calibration-maneuver flight, so we set its single short line aside (Section VI-D), and the map-building flights (Flt1004–1005) are not used. Two honest points remain. First, none of these is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry more attitude variation, which stresses compensation but also aids TL observability by exciting the regressor (Section IV), so performance on a pure operational cruise is not directly established here and is future work. Second, the lines span two altitude regimes: four at the $\approx 400\,\mathrm{m}$ survey-drape altitude and 1007.02 at $\approx 800\,\mathrm{m}$, where upward continuation attenuates the anomaly. Map matching uses the Eastern ("E") and Renfrew ("R") high-resolution anomaly grids, upward-continued to flight altitude with the IGRF core field restored. The metric is horizontal distance-RMS (DRMS) of position error after a $10\,\mathrm{min}$ warm-up. Cold start means the TL coefficients are initialized to zero with no calibration flight. For every comparison the EKF and FGO estimators use identical dynamics, map interpolation, and noise settings (Table I); the window smoother additionally applies the Huber kernel (11), whose isolated contribution is reported separately in Table II ($37.0 \to 32.6\,\mathrm{m}$ on Mag 4), so the reader can attribute the remaining margin to the estimator structure. Unless noted, the window smoother uses $L_w = 5\,\mathrm{min}$ and $L_o = 90\,\mathrm{s}$; L_w was selected on line 1007.06 (Fig. 8) and then held fixed for every other line, so the breadth results of Table III are not per-line tuned. All numbers are produced by continuous-

TABLE I
Estimator settings for all cold-start results; the same $\mathbf{P}_0$, $\mathbf{Q}$, R, and TL settings drive both the EKF and the smoother.

Group	Parameter	Value
Measurement	noise std. $\sqrt{R}$	$12\,\mathrm{nT}$
FOGM S	(σ_S, τ)	$3\,\mathrm{nT}$, $180\,\mathrm{s}$
Initial $\mathbf{P}_0$	pos. / alt. std.	0.1 / $1.0\,\mathrm{m}$
	velocity std. (per axis)	$1.0\,\mathrm{m\,s^{-1}}$
TL prior / walk	$\mathbf{P}_0^\beta$; $\mathbf{Q}^\beta$	$\mathbf{I}(1.0)^2$; $\mathbf{I}(1.0)^2\Delta t$
TL basis	terms (App. B)	P+I+E+bias (19)
Window	L_w / L_o / Δt [s]	300 / 90 / 0.1
Robust	Huber δ	1.345
	IRLS	2nd iter. on, $w \in [10^{-6}, 1]$
Gauss–Newton	max. iter. / tol.	5 / 10^{-4}

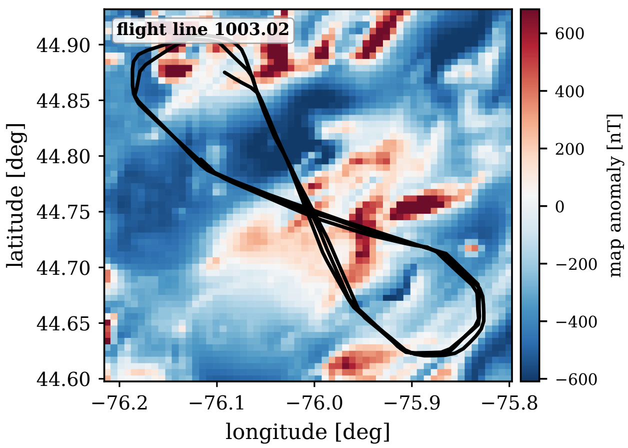

Fig. 6. Line 1003.02 over the Eastern anomaly map.

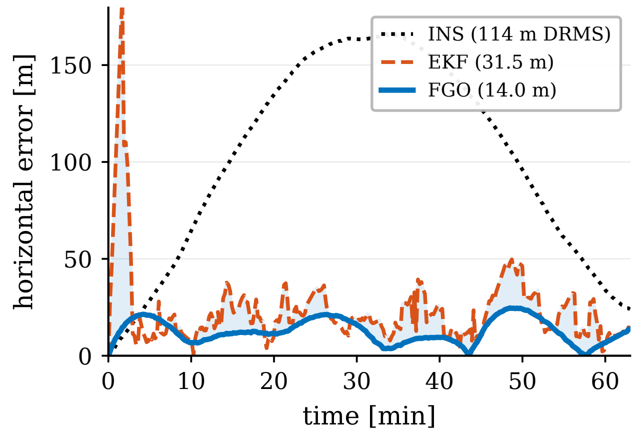

Fig. 7. Horizontal position error on line 1003.02: batch FGO, EKF, and free inertial (DRMS in the legend).

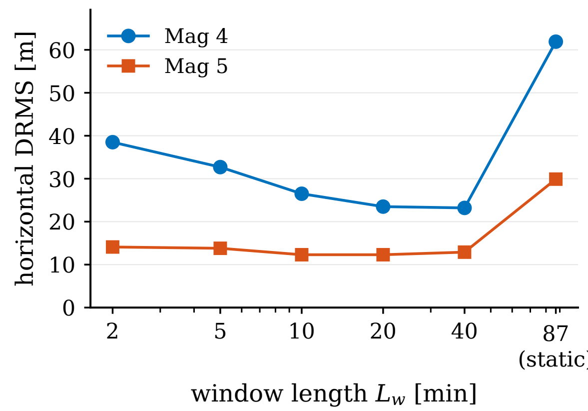

Fig. 8. Window length L_w versus horizontal DRMS on line 1007.06 (cold start).

integration scripts on the code branch and are reproducible from it.

For geographic context, Fig. 6 shows line 1003.02 over the Eastern anomaly field. As a controlled check on the compensated stinger, before the cold-start problem, the batch factor graph tightens the horizontal DRMS from the EKF's $28.3\,\mathrm{m}$ to $14.0\,\mathrm{m}$, against $114.6\,\mathrm{m}$ free-inertial (Fig. 7); its iterated-RTS and sparse-GN/QR solvers agree to within $1\,\mathrm{m}$, an empirical check of Lemma 1.

B. Cold-start line vs. online EKF+TL+NN

The central comparison is the cold start on uncompensated cabin magnetometers, where the interference is present and must be learned online. Table II and Fig. 9 report the primary evaluation line 1007.06 ($87\,\mathrm{min}$, full length) on Mag 4 and Mag 5. A static-TL batch forces a single coefficient vector over the whole line; it underfits the time-varying interference and degrades to $123.7/68.1\,\mathrm{m}$. Allowing the compensation to adapt with a $5\,\mathrm{min}$ window recovers it to $37.0/15.1\,\mathrm{m}$, and the Huber kernel further trims transient outliers to $32.6/14.2\,\mathrm{m}$. Our reimplemented online EKF+TL+NN reaches $40.0/17.5\,\mathrm{m}$ (Section V details how close this is to the reported operating point), while the reported TL-only EKF is weaker still on Mag 4. The window smoother beats the reimplemented NN-augmented filter on both sensors without any neural network, using only the linear TL basis carried as graph variables; against the values reported in [19] it is $4.4\,\mathrm{m}$ better on Mag 4 and $0.2\,\mathrm{m}$ behind on Mag 5. This comparison against the published values is on the single line 1007.06, the only line [19] reports; our reimplemented EKF+TL+NN carries the baseline to the other lines in the breadth study (Table III). We attribute the margin to the window look-ahead: the coefficients are identified from data on both sides of each committed epoch, whereas the EKF must commit with past data only.

The window length itself trades under-determination against staleness. Fig. 8 plots DRMS at the three lengths evaluated on this line: the $2\,\mathrm{min}$ window under-determines the coefficients, the full-line static batch cannot track their drift, and the $5\,\mathrm{min}$ window is the best of the three. The tradeoff mirrors the expressiveness discussion of Section IV and makes L_w a physically interpretable knob rather than a free hyperparameter, though a denser sweep (and other lines, where the balance can shift toward longer windows) is left for the revision.

These cold-start numbers also settle a question the formulation raises: with no neural network, how does the estimator cope with the large fraction of the cabin

TABLE II

Cold start on line 1007.06 (Flt1007), uncompensated cabin magnetometers; DRMS [m]. INS 318 m; compensated-Mag 1 EKF $\approx 20\,\mathrm{m}$.

Method (no NN unless noted)	Mag 4	Mag 5
FGO-online, static batch TL	123.7	68.1
FGO-online, window 2 min	45.9	17.1
FGO-online, window 5 min	37.0	15.1
FGO-online, window 5 min + Huber	**32.6**	**14.2**
EKF+TL+NN, reimplemented [19]	40.0	17.5
EKF+TL+NN, reported [19]	37	14
EKF+TL-only, reported [19]	58	15

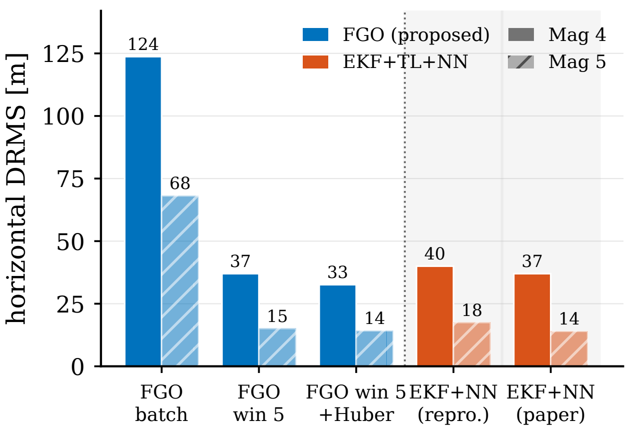

Fig. 9. Cold start on line 1007.06: NN-free fixed-lag window (left) versus the reimplemented and reported EKF+TL+NN (right).

interference that the linear TL basis cannot span. The mechanism is the one built in Section III-A, where the time-varying coefficients, the FOGM disturbance state, and the robust kernel together absorb that remainder as a structured robust regression. The trade it makes is visible in the numbers. The network buys a lower compensation residual, while the window, the disturbance state, and the robust kernel together buy navigation robustness at a higher residual. This is why the uncompensated Mag 4 DRMS of tens to hundreds of metres sits well above survey-grade compensation and still bounds the inertial drift. A learned residual could be added later as an extra factor on the same measurement, for example a Gaussian-process or small-network term, which would place the network's expressiveness inside the graph rather than inside a filter; we leave that to future work.

C. Breadth across lines, flights, and maps

Table III runs the same cold-start TL model on four long lines spanning two flights and two maps, free-flight navigation legs (Flt1007) and survey traverses (Flt1003), grouped by purpose, against three baselines: the plain online-TL EKF, the reimplemented EKF+TL+NN of Section V (the strong causal method that a fair comparison demands), and a marginalized (Rao-Blackwellized) particle filter extended to carry the TL coefficients in its conditionally-linear-Gaussian block (MPF+TL), a genuine non-Gaussian Bayesian estimator rather than a linearization.

The baselines behave very differently, and the contrast is the point. The plain online-TL EKF is fragile: on the noisy Mag 4 it diverges from the cold start to tens of kilometres on two lines and runs off the map on a third. The particle filter is no more robust—it diverges past $10\,\mathrm{km}$ on every counted case, because per-particle map-matching still cannot re-linearize the early flight once the coefficients drift. The neural network is what fixes this: the EKF+TL+NN stays finite on all eight cases, confirming that its residual network is what buys cold-start robustness for the causal filter. The window smoother matches that robustness without any neural network (it is bounded on all eight) and improves on accuracy: across the four full-length lines ($63\,\mathrm{min}$ to $87\,\mathrm{min}$, eight mag-cases) it is the best of the four on all eight, beating the NN-augmented filter by $4\,\mathrm{m}$ to $91\,\mathrm{m}$ (Fig. 10).

We also ran the short calibration-maneuver line 1006.08 (Flt1006) but set it aside rather than tabulate it: it is only $14\,\mathrm{min}$ long, so the $10\,\mathrm{min}$ warm-up leaves a $4\,\mathrm{min}$ window over which none of the DRMS values is reliable, and its low map-information floor (the Remark in Section IV) makes every method weak. It reproduces the same causal-EKF divergence, but we draw no accuracy conclusion from it: in particular the NN's $108.3\,\mathrm{m}$ on its Mag 5, against $117\,\mathrm{m}$ to $122\,\mathrm{m}$ for the TL-only EKF and window, is not evidence of an NN accuracy advantage on a window this short.

Two effects in the reliable cases are worth naming. First, the EKF+TL+NN follows the reproduced design in feeding the network only the permanent TL group, whereas the online-TL EKF and the window smoother both carry the full permanent-induced-eddy-bias basis; on the well-conditioned Mag 5 those linear higher-order terms matter, so the NN filter can fall behind even the plain EKF (1003.02: 29.5 vs $28.1\,\mathrm{m}$) when its network has not yet learned what the dropped linear terms would have supplied directly. Second, and relatedly, the network's extra online-learned states add estimation variance where the linear basis already suffices, a cost the NN-free window avoids. Both effects favor a linear compensation carried in a smoother over a learned one carried in a filter, on this benchmark. One caveat on provenance: the EKF+TL+NN is our reimplementation, matching the reported 1007.06 operating point to within a few metres (Section V), but the original's per-line numbers on the other lines are unpublished, so the comparison is against a faithful, not the authors', filter. Its cold-start trajectory is also numerically sensitive, so its per-line DRMS varies by order 1–$2\,\mathrm{m}$ between runs; the values here are those of the committed continuous-integration run and do not affect the best-of-four ordering.

Two conclusions follow, and they are worth separating because a single-line comparison conflates them. The first is about robustness. Keeping a causal filter bounded at cold start is not unique to the window, since the neural

TABLE III

Breadth: cold-start cabin magnetometers, DRMS [m]. Purpose FF = free-flight (Flt1007), SV = survey (Flt1003); “div.” $> 10\,\mathrm{km}$, “err.” = off map; bold = best of four. MPF+TL is a Rao-Blackwellized particle filter carrying the TL coefficients; it diverges on every counted case. Calibration line 1006.08 set aside (see text).

Line	Purp.	Map	alt [km]	Mag	EKF online	EKF+ TL+NN	MPF +TL	FGO win
1007.06	FF	R	0.4	4	46.7	48.9	div.	**32.7**
1007.06	FF	R	0.4	5	17.8	18.3	div.	**13.8**
1007.02	FF	E	0.8	4	div.	130.0	div.	**38.6**
1007.02	FF	E	0.8	5	31.6	30.4	div.	**14.5**
1003.02	SV	E	0.4	4	div.	99.1	div.	**42.6**
1003.02	SV	E	0.4	5	28.1	29.5	div.	**21.7**
1003.08	SV	R	0.4	4	err.	46.6	div.	**26.1**
1003.08	SV	R	0.4	5	21.1	20.7	div.	**12.4**
FGO best (all counted lines)								**8 / 8**

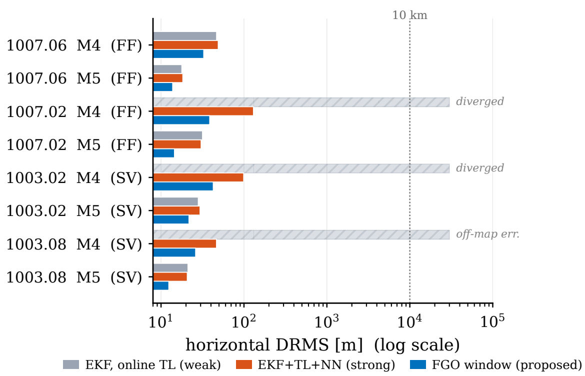

Fig. 10. Breadth study (Table III), log DRMS axis: online-TL EKF (gray), EKF+TL+NN (red), and the NN-free FGO window (blue). The MPF+TL particle filter diverges on every counted case and is omitted.

network does it too, so the blanket claim that causal filters diverge is false for the state-of-the-art filter. It is, however, true of every non-NN cold-start estimator we tried, including the marginalized particle filter, which diverges on all eight counted cases despite being a full Bayesian estimator: bounded cold-start compensation without a network requires the batch/window re-smoothing, not merely a non-Gaussian representation. An ablation within our own estimator locates the mechanism more precisely: running the same window with the robust kernel disabled still stays bounded on all eight counted cases (e.g. $66.3\,\mathrm{m}$ on 1003.02/Mag 4 where the causal filter reads $41\,626\,\mathrm{m}$, refined to $42.6\,\mathrm{m}$ by Huber), so the robustness comes from the windowed relinearization of Section III-A, and the kernel buys accuracy, not survival. The recorded platform currents let us check the absorption account of Section III-A directly on line 1003.02: the fourteen onboard current channels linearly explain $28\,\%$ to $39\,\%$ of the raw cabin interference (strobe lights, cabin fuel pump, and INS heater lead), but only $2\,\%$ to $6\,\%$ of the post-fit windowed-FGO residual, so the current-aligned component is almost entirely absorbed by the time-varying $\boldsymbol{\beta}_t$ and S_t; and the Huber-down-weighted epochs concentrate 2.6–

TABLE IV

Estimator families for cold-start map-matching compensation. ✓ = yes, ✗ = no, (∼) = limited.

Estimator	Future data	Comp. state	Non-Gauss.	Bounded latency
Causal EKF [4]	✗	(∼)	✗	✓
Particle / grid filter [8]	✗	✗	✓	✓
Online EKF+TL+NN [19]	✗	✓	(∼)	✓
Batch FGO	✓	✓	✓	✗
Proposed (fixed-lag FGO)	✓	✓	✓	✓

$3.6\times$ on current-switching activity, with strobe switching alone correlating at 0.55 with the residual magnitude on Mag 5, confirming that the kernel attenuates precisely the transients the linear states cannot track. These are instantaneous linear regressions, hence lower bounds on the current contribution. What the window adds is that it reaches the same bounded regime without the network, its online-training machinery, or the verification burden that comes with it. The second is about accuracy, where the window is the best of the four on all eight full-length cases. This margin is real, but it should be read with care, because our EKF+TL+NN is a reimplementation and part of its deficit is that its online-learned states add variance on the well-conditioned sensor, where the linear TL basis already suffices. The defensible reading is that a linear estimator with no network, given future context through the window, is enough both to survive the cold start and to match or beat an NN-augmented causal filter on this benchmark. Table IV places the estimator families along the axes that matter for cold-start compensation.

The observability condition of Section IV accounts for the pattern. The endogeneity cancellation (Remark 1) is why the scalar magnetometer had to be excluded from the network inputs for the baseline to converge, and exogeneity (Corollary 1) is why the attitude-only TL basis stays safe at any dimension. The map-information floor of the Remark is why the short calibration line 1006.08 is hard for every method, since there is little independent map gradient to exploit over its brief window, and it is the reason we set that line aside. The condition is geometric and predictive only through these qualitative rules; a scalar screening statistic we tried did not generalize, which we report as a negative result.

D. Sensor-error factors (simulation)

To exercise the sensor-error factors of Section III we inject the optically-pumped-magnetometer error model of [20], [21] into a simulated flight (free-inertial reference $31.1\,\mathrm{m}$): a cosine-harmonic heading error at orders $\{1, 2, 4\}$ ($7.6\,\mathrm{nT}$ RMS), a fluxgate hard-iron bias, a linear drift, and the equatorial-dead-zone noise amplification $1/\max(|\cos\psi|, \epsilon)$. We then enable the corresponding graph variables cumulatively; Table V reports the result. With no sensor-error states the estimate is worse than free inertial ($42.9\,\mathrm{m}$): the unmodeled error corrupts the map

TABLE V
Sensor-error factors (simulation), enabled cumulatively; DRMS [m]. Free-inertial 31.1 m; non-robust solver (Huber within 2 m).

Graph variables enabled	DRMS
none (map aiding only)	42.9
+ heading harmonics $\{1, 2, 4\}$	18.9
+ dead-zone weight	15.5
+ hard-iron bias	8.1
+ **linear drift (full)**	**6.2**

aiding, exactly the mechanism of Proposition 1. Enabling all four factors reduces the error to 6.2 m, an 86 % reduction, monotone in this order. Because the factors are enabled cumulatively in a single order, Table V attributes marginal gains only for that order: the largest single-step drop comes from the heading harmonics (42.9 → 18.9 m), followed by the dead-zone weighting (→ 15.5 m, now load-bearing because the injected dead-zone noise is heteroscedastic), the hard-iron bias (→ 8.1 m), and the linear drift, whose rate the estimator recovers cleanly ($\hat{\gamma}_1 = 0.018$ versus a true 0.020). The heading coefficients themselves are poorly identified ($\hat{c}_1 = 3.8$ versus a true 10): level flight sweeps the sensor–field angle by only about 12°, so the harmonic regressor is barely excited, an observability property of the trajectory, consistent with the Remark of Section IV, that a maneuvering flight would relieve, and the gain to navigation comes from the fitted combination over the observed angle range, not from the individual coefficients. The Huber kernel gives similar values (within 2 m; it helps most at the dead-zone step). Two caveats bound this study: it is simulation-only, and the injected errors share the estimator's functional form, so it demonstrates that the graph formulation can carry and recover such states, not that these models capture real sensor behavior.

E. Consistency and statistical significance

Each real line provides a single physical realization, so its DRMS is a point estimate and its measurement noise cannot be resampled. To assess statistical significance and, more importantly, whether the reported covariance is credible, we run a Monte-Carlo study in simulation, where repeated realizations are valid. We fix one simulated trajectory over the high-resolution Eastern anomaly map and, for each of 30 seeds, draw an independent INS-error realization and an independent clean scalar-measurement realization, then run three map-matching estimators on the same draw: the causal EKF, the marginalized (Rao-Blackwellized) particle filter already in the toolbox, and the batch factor graph. The clean compensated measurement isolates the navigation covariance from the cold-start compensation transient.

Averaged over the 30 seeds, after a 30 s warm-up, the horizontal DRMS is 1.4 m for the factor graph, 4.1 m for the EKF, and 4.9 m for the particle filter, with 95 % confidence half-widths of 0.2, 0.4, and 0.8 m. Even on this

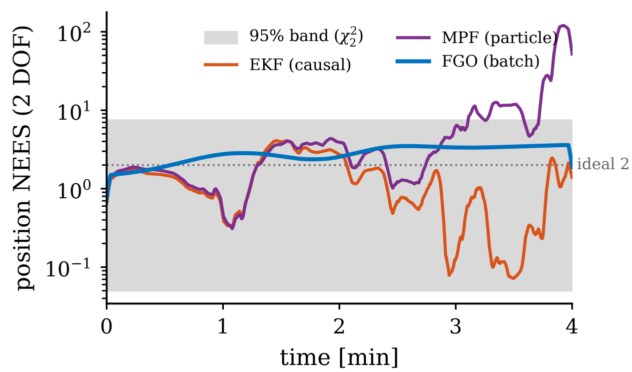

Fig. 11. Monte-Carlo position NEES versus time (EKF, marginalized particle filter, and batch FGO) with the 95 % χ^2_2 band; the EKF and FGO stay inside it while the particle filter sits far above (ideal 2, dotted).

clean sensor the factor graph separates cleanly from both recursive baselines—its interval does not overlap theirs—while the EKF and the particle filter are statistically indistinguishable in accuracy. The larger cold-start separations are in Sections VI-C and VI-D; the purpose of this study is consistency.

For consistency we use the per-axis two-degree-of-freedom position NEES, $(e_N/\sigma_N)^2 + (e_E/\sigma_E)^2$, whose expectation under a consistent covariance is 2. Averaged over all seeds and epochs the ANEES is 1.57 for the factor graph and 1.72 for the EKF, both below the ideal 2: both report conservative uncertainty that bounds the true error with margin. The particle filter is the opposite. Its ANEES is 59.8, more than an order of magnitude above 2, so its reported position covariance is severely over-confident—the particle-depletion signature, in which the surviving particles after resampling on a high-resolution map cluster far more tightly than the true error and the empirical covariance collapses. The two linearized estimators are credible; the particle filter, though DRMS-competitive with the EKF, is not. Fig. 11 shows the EKF and factor-graph NEES staying inside the 95 % χ^2_2 band throughout a representative run while the particle filter runs far above it, and Fig. 12 shows the North and East position error contained within the factor-graph $\pm 2\sigma$ envelope. Two caveats bound the reading. The NEES uses the per-axis (diagonal) covariance, since the cross term is not exposed by the evaluated filter output, and the study is a matched-model simulation, so it certifies internal consistency rather than robustness to model mismatch. The conservative bias of the linearized estimators indicates that their process noise could be tightened, which is the safe direction in which to err.

F. Computational considerations

Each window is a sparse nonlinear least-squares problem of fixed size $O\big(L_w(n+m)\big)$; the block-bidiagonal Jacobian of the two chains (Fig. 3) is factored in $O(L_w)$ by the square-root solver [35], and the number of Gauss–Newton iterations is small (typically 3–5) because the

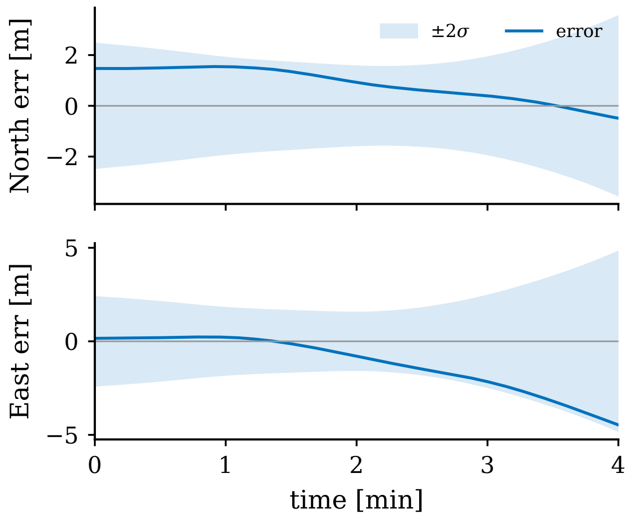

Fig. 12. North and East position error within the estimator $\pm 2\sigma$ bands (representative simulated FGO run).

problem is mildly nonlinear away from observability collapse. The per-window cost is therefore constant in the line length and the overall complexity is linear in the number of epochs, matching a filter's asymptotic cost while retaining the look-ahead. Memory is bounded by the window, not the flight. The one-time overhead relative to an EKF is the window factorization and the IRLS reweighting, which in our unoptimized research implementation runs comfortably faster than real time on the SGL sampling rate.

These same properties make the estimator compatible with real-time operation. Its committed output lags real time by the overlap L_o, here $90\,\mathrm{s}$, a latency that suits the map-aiding role, in which the INS supplies the high-rate solution and the smoother corrects it periodically, and that the designer trades against accuracy through L_o. The factor interface is modular, so the compensation, sensor-error, and map factors can be added or removed without touching the solver, and one implementation then covers the calibrated case with frozen coefficients, the cold-start case with adaptive coefficients, and the augmented-sensor case. Integrating it with an operational system amounts to feeding the INS mechanization, fluxgate, and scalar magnetometer streams into the graph and reading back the corrected position at the commit rate. The main tuning burden is the two window parameters and the compensation process noise $\mathbf{Q}^{\beta}$, all of which have clear physical meaning.

VII. Conclusion

This paper formulated airborne magnetic-anomaly navigation as factor-graph MAP estimation and used the formulation to solve the cold-start compensation problem jointly with navigation. The Tolles–Lawson coefficients enter the graph as time-varying variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight and late-arriving calibration information corrects early navigation states, the mechanism a one-pass causal filter lacks; physically modeled scalar-magnetometer error terms (heading harmonics, hard-iron bias, drift, and dead-zone reweighting) attach to the same graph as ordinary variables; and a necessary-and-sufficient joint-observability condition, with its exogeneity rule and endogeneity cancellation, states when position and compensation can be separated at all. On real SGL 2020 data the estimator stays bounded from a cold start on all eight full-length cabin-magnetometer cases, where both the plain online-TL EKF and a marginalized particle filter diverge, and it is more accurate on every such case than a reimplemented online EKF+TL+NN, with no neural network and no calibration flight. A simulation Monte-Carlo shows its reported covariance is consistent and conservative where the particle filter's is over-confident, and a cumulative ablation shows the physical sensor-error factors recover most of an injected sensor-error budget.

Several limitations bound the study. Each real line is a single physical realization, so its DRMS is a point estimate; the Monte-Carlo supplies distributional statements in simulation only. The baseline set, though it now spans a weak causal filter, the strongest reported causal method, and a particle filter, still lacks a grid or point-mass MMSE map-matcher [23], [8]. The sensor-error factors are validated against injected, model-matched errors in simulation, not against a physical optically-pumped magnetometer. And none of the SGL 2020 lines is a long, straight-and-level operational transit: the attitude content that stresses compensation also excites the TL regressor and aids its observability, so performance under the weak excitation of a pure cruise is not established here. Short lines remain hard for every method, an information limit rather than an estimator one.

These limitations set the agenda for further work: extending the evaluation to the full SGL line set and a grid or point-mass baseline; demonstrating the sensor-error factors on flight data with genuine optically-pumped-magnetometer heading error, with maneuvers wide enough to make the harmonic coefficients cleanly identifiable; testing a low-excitation operational cruise; and sharpening the geometric observability condition into a quantitative identifiability screen. None of these qualifications affects the central and reproducible finding: carrying the compensation as time-varying variables of a windowed factor graph is sufficient to survive the cold start and to match or beat an NN-augmented causal filter on this benchmark, with every estimated quantity a physically interpretable state.

Appendix A
Pinson Error-State Dynamics

The navigation-error block of $\mathbf{\Phi}_t$ in (2) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors $(\delta\mathbf{p}, \delta\mathbf{v}, \boldsymbol{\psi})$ and sensor biases $(\mathbf{b}_a, \mathbf{b}_g)$, the continuous error dynamics are

$$\begin{aligned} \dot{\delta\mathbf{p}} &= \mathbf{F}_{pp}\,\delta\mathbf{p} + \delta\mathbf{v}, \\ \dot{\delta\mathbf{v}} &= \mathbf{F}_{vp}\,\delta\mathbf{p} + \mathbf{F}_{vv}\,\delta\mathbf{v} - [\mathbf{f}^n\times]\,\boldsymbol{\psi} + \mathbf{C}_b^n\mathbf{b}_a, \\ \dot{\boldsymbol{\psi}} &= \mathbf{F}_{\psi p}\,\delta\mathbf{p} + \mathbf{F}_{\psi v}\,\delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n\times]\,\boldsymbol{\psi} - \mathbf{C}_b^n\mathbf{b}_g, \end{aligned} \tag{13}$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, $\boldsymbol{\omega}_{in}^n$ the inertial rate, $[\cdot\times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ. The discrete $\boldsymbol{\Phi}_t = \exp(\mathbf{F}_t\,\Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [46]. Only $\mathbf{g}_t^\top$, the position row of the measurement Jacobian (10), couples this block to the map.

Appendix B
Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of (6) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [\,u\ v\ w\,]^\top$ (with $u^2 + v^2 + w^2 = 1$) and the scalar field magnitude B [11], [12]. The permanent moment contributes the three cosines (u, v, w); the induced magnetization, symmetric in the field, contributes the six products $B\,(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B\,(u\dot{u}, u\dot{v}, \ldots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top\boldsymbol{\beta}_t$ is linear in the coefficients $\boldsymbol{\beta}_t$, the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes $\boldsymbol{\beta}$ is unavailable, which is precisely why we carry it as a graph variable.

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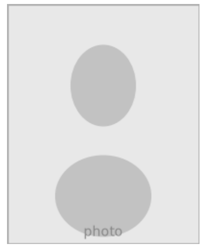

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